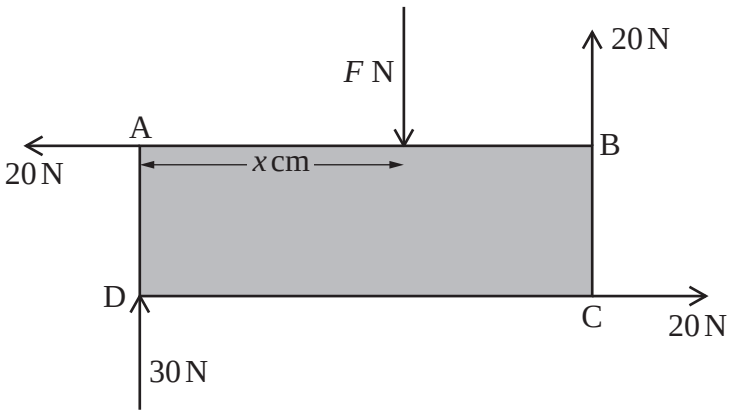
- The acceleration due to gravity is denoted by $g\,\text{m s}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.

**5** ABCD is a rectangular lamina in which AB is 30 cm and AD is 10 cm.

Three forces of 20 N and one force of 30 N act along the sides of the lamina as shown in the diagram.




An additional force $F$ N is also applied at right angles to AB to a point on the edge AB $x$ cm from A.

**(a)** Given that the lamina is in equilibrium, calculate the values of $F$ and $x$. **[3]**

The point of application of the force $F$ N is now moved to B, but the magnitude and direction of the force remain the same.

**(b)** Explain the effect of the new system of forces on the lamina. **[2]**